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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x \sin 2x}{1 - \cos 3x} &= \frac{0}{0} \\&= \lim_{x \rightarrow 0} \frac{\sin 2x \cdot 2 \cdot x^2}{2x \cdot (1 - \cos 3x)} \\&= \lim_{x \rightarrow 0} \frac{2x^2}{(1 - \cos 3x)} \\&= \lim_{x \rightarrow 0} \frac{2x^2 \cdot (1 + \cos 3x)}{(1 - \cos 3x) \cdot (1 + \cos 3x)} \\&= \lim_{x \rightarrow 0} \frac{2x^2 \cdot (1 + \cos 3x)}{\sin^2 3x} \\&= \left(\lim_{x \rightarrow 0} \frac{9x^2}{\sin^2 3x} \right) \cdot \lim_{x \rightarrow 0} (1 + \cos 3x) \cdot \frac{2}{9} \\&= \left(\lim_{x \rightarrow 0} \frac{3x}{\sin 3x} \right)^2 \cdot \frac{4}{9} \\&= 1 \cdot \frac{4}{9} \\&= \frac{4}{9}\end{aligned}$$

References